Exercise J01

Q1. Let f(x) = |x – 1|, then -

(A) $f(x^2) = (f(x))^2$ (B) f(x+y)= f(x) + f(y) (C) f(|x|) = |f(x)| (D) none of these

Ans. D

Sol. -

Q2. If $f(x) = \cos\left[\frac{\pi^2}{2}\right]x + \sin\left[-\frac{\pi^2}{2}\right]x$, where [x] denotes the greatest integer function, then which of the following is not correct –

(A) f(0) = 1 (B) $f\left(\frac{\pi}{3}\right) = \frac{1}{\sqrt{3}+1}$ (C) $f\left(\frac{\pi}{2}\right) = 0$ (D) $f(\pi) = 0$

Ans. D

Sol. -

Q3. Let f(x) be a function whose domain is [– 5, 7]. Let g(x) = |2x + 5|. Then domain of (fog) (x) is

(A) [– 4, 1] (B) [– 5, 1] (C) [– 6, 1] (D) none of these

Ans. C

Sol. Domain of f(g(x)). Range of g(x) ≡ Domain of f(x)

$\Rightarrow \quad -5 \le |2x + 5| \le 7 \qquad \Rightarrow \quad 0 \le |2x+5| \le 7 \qquad \Rightarrow \quad -7 \le 2x + 5 \le 7$

$\Rightarrow \quad -12 \le 2x \le 2 \qquad \Rightarrow \quad -6 \le x \le 1$

Q4. The domain of the function $f(x) = \log_{1/2}\left(-log_2\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right)$ is:

(A) 0 < x < 1 (B) 0 < x ≤ 1 (C) x ≥ 1 (D) null set

Ans. D

Sol. $f(x) = \log_{1/2}\left(-log_2\left(1+\frac{1}{4\sqrt{x}}\right)-1\right) \quad \Rightarrow \quad -\log_2\left(1+\frac{1}{x^{1/4}}\right)-1>0$

$\Rightarrow -\infty < \log_2 < -1 \Rightarrow 0 < 1 + \frac{1}{x^{1/4}} < \frac{1}{2} \quad \Rightarrow \quad -1 < \frac{1}{x^{1/4}} < -\frac{1}{2}$

$\Rightarrow x \in \phi$ (null set)

Q5. The domain of the function , $f(x) = \frac{sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is :

(A) $[1, 2]$ (B) $[2, 3)$ (C) $[2, 3]$ (D) $[1, 2)$

Ans. B

Sol. To define || (x), $9 - x^2 > 0 \Rightarrow -3 < x < 3$
..................(1)

$-1 \leq (x-3) \leq 1 \Rightarrow 2 \leq x \leq 4$
...................(2)

From (i) and (ii), $2 \leq x < 3$ i.e., $[2, 3)$

Q6. The domain of the function $f(x) = \dfrac{1}{\sqrt{|x| - x}}$ is

(A) $(-\infty, \infty)$ (B) $(0, \infty)$ (C) $(-\infty, 0)$ (D) $(-\infty, \infty) - \{0\}$

Ans. C

Sol. $f(x) = \dfrac{1}{\sqrt{|x| - x}}$

For domain of real function

$|x| - x > 0$

$|x| > x$

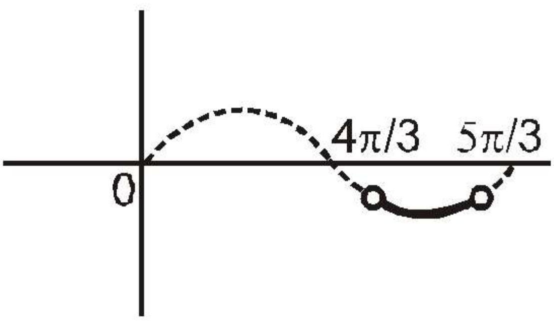
$x \in (-\infty, 0)$

Q7. If $[2 \cos x] + [\sin x] = -3$, then the range of the function, $f(x) = \sin x + \sqrt{3} \cos x$ in $[0, 2\pi]$ is:
(where [.] denotes greatest integer function)

(A) $[-2, -1)$ (B) $(-2, -1]$ (C) $(-2, -1)$ (D) $[-2, -\sqrt{3})$

Ans. D

Sol.



$[2 \cos x] + [\sin x] = -3$

$-2 \leq 2 \cos x \leq 2 \quad \Rightarrow \quad [2 \cos x] = 2, 1, -1, -2$

$-1 \leq \sin x \leq 1 \quad \Rightarrow \quad [\sin x] = 1, 0, -1$

Equation holds true for $[2 \cos x] = -2$ and $[\sin x] = -1$

$\Rightarrow\ -1 \le \cos\ x < -\frac{1}{2}$ and $-1 \le \sin x < 0$

$\Rightarrow\ x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$ and $x \in (\pi, 2\pi)$ $x \in \left(\pi, \frac{4\pi}{3}\right)$

$f(x) = \sin x + \sqrt{3} \cos x$

$f(x) = 2 \sin\left(x + \frac{\pi}{3}\right)$

$\frac{4\pi}{3} < x + \frac{\pi}{3} < \frac{5\pi}{3}$ $\Rightarrow$ $-1 \le \sin\left(x + \frac{\pi}{3}\right) < -\frac{\sqrt{3}}{2}$ $\Rightarrow$ $-2 \le 2$

$\sin\left(x + \frac{\pi}{3}\right) < -\sqrt{3}$

Hence range is $[-2, -\sqrt{3}]$

Q8. Range of $f(x) = \log(3x^2 - 4x + 5)$ is

(A) $\displaystyle \bigg[\log \frac{11}{3},\infty\bigg)$ (B) $[\log 10 , \infty)$

(C) $\displaystyle \bigg[\log \frac{11}{6},\infty\bigg)$

(D) $\displaystyle \bigg[\log \frac{11}{12},\infty\bigg)$

Ans. A

Sol. $f(x) = \log_e(3x^2 - 4x + 5)$

$3x^2 - 4x + 5 \ge \displaystyle \frac{11}{3}$ $\Rightarrow$ $ln(3x^2 - 4x + 5) \ge ln \displaystyle \frac{11}{3}$ [$\because$ ln is an increasing function]

$\therefore$ Range is $\displaystyle \bigg[\ln \frac{11}{3},\infty\bigg)$

Q9. Range of $f(x) = 4^x + 2^x + 1$ is

(A) $(0, \infty)$ (B) $(1, \infty)$ (C) $(2, \infty)$ (D) $(3, \infty)$

Ans. B

Sol. $f(x) = 4^x + 2^x + 1$

Let $2^x = y > 0$, $\forall$ x $\in$ R

$\therefore$ $f(x) = g(t) = t^2 + t + 1,\ t > 0$

$g(t) = \displaystyle \bigg(t+\frac{1}{2}\bigg)^2 + \displaystyle \frac{3}{4}$ $\Rightarrow$ $\displaystyle \bigg(t+\frac{1}{2}\bigg) > \displaystyle \frac{1}{2}$ $\Rightarrow$ $\displaystyle \bigg(t+\frac{1}{2}\bigg)^2 > \displaystyle \frac{1}{4}$ $\Rightarrow$ $\displaystyle \bigg(t+\frac{1}{2}\bigg)^2 + \displaystyle \frac{3}{4} > 1$

Range is $(1, \infty)$

Q10. The range of the functions $f(x) = \displaystyle \log_{\sqrt2}\bigg(2-\log_2\bigg(16\sin^2x+1\bigg)\bigg)$ is

(A) $(-\infty, 1)$ (B) $(-\infty, 2)$ (C) $(-\infty, 1]$ (D) $(-\infty, 2]$

Ans. D

Sol. Here $(2 - \log_2)(16 \sin^2 x + 1) > 0$ $\Rightarrow$ $0 < 16 \sin^2 x + 1 < 4$

$\Rightarrow$ $\displaystyle \frac{-1}{16} \le \sin^2 x < \displaystyle \frac{16}{3}$

$\Rightarrow$ $1\ 16 \sin^2 x + 1 \le 4$ $\Rightarrow$ $0 \le \log_2 (16 \sin^2 x + 1) < 2$

$\Rightarrow$ $2 \ge 2 - \log_2 (16 \sin^2 x + 1) > 0$

$\Rightarrow$ $\displaystyle \log_{\sqrt{2}} 2 \ge \displaystyle \log_{\sqrt{2}} (2 - \log_2 (16 \sin^2 x + 1) > \infty$ $\Rightarrow$ $2 \ge y > \infty$

Hence range is $y \in (-\infty, 2]$

Q11. The range of the function, $f(x) = \displaystyle \frac{2+x}{2-x}$, $x \ne 2$ is :

(A) R (B) $R - \{-1\}$ (C) $R - \{1\}$ (D) $R - \{2\}$

Ans. B

Sol. $\displaystyle f\left(x \right) = \frac{{linear}}{{linear}}$

$\displaystyle f\left(x \right) = \frac{{2 + x}}{{2 - x}},x \ne 2$

$\displaystyle y \in R - \left\{ {\frac{1}{{ - 1}}} \right\}$

$y \in R - \{-1\}$

Q12. The range of the function, $f(x) = {}^{7-x}P_{x-3}$ is :

(A) $\{1, 2, 3, 4\}$ (B) $\{1, 2, 3, 4, 5, 6\}$ (C) $\{1, 2, 3, 4, 5\}$ (D) $\{1, 2, 3\}$

Ans. D

Sol. $|(3) = {}^{7-3}P_0 = 1$, $|(4) = {}^{3}P_1 = 3$ and $|(5) = {}^{2}P_2 = 2$

Hence, range of $| = \{1, 2, 3\}$

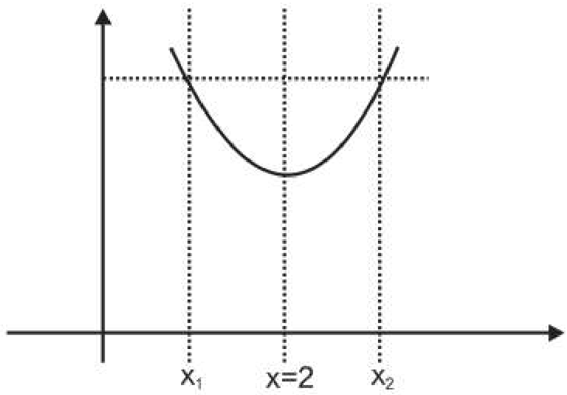
Q13. The graph of the function, $y = f(x)$ is symmetrical about the line $x = 2$, then :

(A) $f(x) = f(-x)$ (B) $f(2 + x) = f(2 - x)$ (C) $f(x + 2) = f(x - 2)$

(D) $f(x) = -f(-x)$

Ans. B

Sol. Let us consider a graph symmetric w.r.t. line $x = 2$ as shown in figure



from figure $f(x_1) = f(x_2)$ where $x_1 = 2 - x$ & $x_2 = 2 + x$ $\therefore$ $f(2 - x) = f(2 + x)$

Q14. If $f(x) = \displaystyle \frac{4a-7}{3} x^3 + (a - 3) x^2 + x + 5$ is a one-one function, then

(A) $2 \le a \le 8$ (B) $1 \le a \le 2$ (C) $0 \le a \le 1$ (D) None of these

Ans. A

Sol. -

Q15. A function ' f ' from the set of natural numbers to integers defined by ,

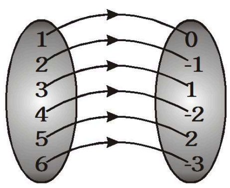
$$f(x) = \begin{cases} \frac{n-1}{2} & \text{, when 'n' is odd} \\ -\frac{n}{2} & \text{, when 'n' is even} \end{cases}$$ is :

(A) onto but not one-one (B) one-one and onto both (C) one-one but not onto

(D) neither one-one nor onto

Ans. B

Sol. | : N $\to$ I



| (1) = 0, |(2) = – 1, (3) = 1, | (4) = – 2, | (5) = 2 and | (6) = – 3 so on.

In this type of function every elements of set A has unique image in set B and there is no element left in set B. Hence I is one-one and onto function.

Q16. If $f : R \to S$, defined by $f(x) = \sin x - \displaystyle \sqrt3 \cos x + 1$, is onto , then the interval of ' S ' is :

(A) $[0, 1]$ (B) $[-1, 1]$ (C) $[0, 3]$ (D) $[-1, 3]$

Ans. D

Sol. Using $-\displaystyle \sqrt{a^2+b^2}$ (a sin x + b cos x) $\le$ $\displaystyle \sqrt{a^2+b^2}$

$-\displaystyle \sqrt{1+(-\sqrt{3})^2}$ $\le$ (sin x $-$ $\displaystyle \sqrt{3}$) $\le$ $\displaystyle \sqrt{1+(-\sqrt{3})^2}$

-2 $\le$ (sin x $-$ $\displaystyle \sqrt{3}$ cos x) $\le$ 2

$-2 + 1$ $\le$ (sin x $-$ $\displaystyle \sqrt{3}$ cos x $+1$) $\le$ $2 + 1$

-1 $\le$ (sin x $-$ $\displaystyle \sqrt{3}$ cos x + 1) $\le$ 3 i.e., range = [–1, 3]

$\therefore$ For | to be onto S = [–1, 3]

Q17. For real x, let $f(x) = x^3 + 5x + 1$, then -

(A) f is one – one but not onto R (B) f is onto R but not one – one

(C) f is one – one and onto R (D) f is neither one – one nor onto R

Ans. C

Sol. For real x, $f(x) = x^3 + 5x + 1$

$\displaystyle \lim\limits_{x \to \infty}$ f(x) = + ∞

and $\displaystyle \lim\limits_{x \to \infty}$ f(x) = – ∞

$\therefore$ Range is R – f(x)

Now $f'(x) = 3x^2 + 5 > 0$

$\therefore$ f(x) is one-one

f(x) is one-one onto

Q18. The largest interval lying in $\displaystyle \bigg(-\frac{\pi}{2},\frac{\pi}{2}\bigg)$ for which the function ,
$\displaystyle \bigg[f{x} = 4^{-x^2}+\cos^{-1}\bigg(\frac{x}{2}-1\bigg)+log(\cos x)\bigg]$ is defined as :

(A) $\displaystyle \bigg(-\frac{\pi}{2},\frac{\pi}{2}\bigg)$

(B) $\displaystyle \bigg[-\frac{\pi}{2},\frac{\pi}{2}\bigg)$ (C) $\displaystyle \bigg[0\,,\frac{\pi}{2}\bigg)$

(D) [0, π]

Ans. C

Sol. f(x) is defined if – 1 $\le$ $\displaystyle \frac{\pi}{2}$ – 1 $\le$ 1 and cos x > 0

or 0 $\le$ x $\le$ 4 and – $\displaystyle \frac{\pi}{2}$ < x < $\displaystyle \frac{\pi}{2}$

$\therefore$ x $\in$ $\displaystyle \bigg[0,\frac{\pi}{2}\bigg)$

Q19. If $f(x) = \cos(\ln x)$ then, $f(x)\,f(y) - \displaystyle\frac{1}{2}\bigg[f\bigg(\frac{x}{y}\bigg)+f(xy)\bigg]$ has the value

(A) -1 (B) $\displaystyle\frac{1}{2}$ (C) -2 (D) None of these

Ans. D

Sol. -

Q20. If $y = f(x)$ satisfies the condition $f\displaystyle\bigg(x+\frac{1}{x}\bigg) = x^2 + \frac{1}{x^2}$ $(x \ne 0)$ then $f(x) =$

(A) $-x^2 + 2$ (B) $-x^2 + 2$ (C) $x^2 - 2,\ x \in R - \{0\}$ (D) $x^2 - 2,\ |x| \in [2, \infty)$

Ans. D

Sol. -

Q21. If $f(1) = 1$ and $f(n + 1) = 2f(n) + 1$ if $n \ge 1$, then $f(n)$ is equal to

(A) $2^n + 1$ (B) 2^n (C) $2^n - 1$ (D) $2^{n-1} - 1$

Ans. C

Sol. -

Q22. If $f : R \to R$ satisfies $f(x + y) = f(x) + f(y)$, for all $x, y \in R$ and $f(1) = 7$, then $\displaystyle\sum\limits_{r=1}^n f(r)$ is :

(A) $\displaystyle\frac{7(n+1)}{2}$ (B) $7n(n + 1)$ (C) $\displaystyle\frac{7n(n+1)}{2}$

(D) $\displaystyle\frac{7n}{2}$

Ans. C

Sol. $f(x + y) = f(x) + f(y) \quad \forall x, y \in R$

$\Rightarrow f(x) = kx$

Given $f(1) = 7$

$\Rightarrow k = 7$

$\displaystyle \Rightarrow \sum\limits_{r = 1}^n {f\left(r \right)} = \sum\limits_{r = 1}^n {7r = \frac{{7n\left({n + 1} \right)}}{2}}$

Q23. If $f(x) + 2f\displaystyle\bigg(\frac{1}{x}\bigg) = 3x$, $x \ne 0$ and $S = \{x \in R : f(x) = f(-x)\}$; then S :

(A) contains exactly two elements (B) contains more than two elements

(C) is an empty set (D) contains exactly one element

Ans. A

Sol. $f(x) + 2f(1/x) = 3x$ ....(1)

$f(f/x) + 2f(x) = \displaystyle \frac{3}{x}$ ....
(2)

From (1) & (2)

$f(x) = f(-x)$

$x^2 = (2)$

$x = \pm \displaystyle \sqrt{2}$

S contains exactly two elements

Q24. Let a, b, c $\in$ R. If $f(x) = ax^2 + bx + c$ is such that $a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy$, $\forall$ x, y $\in$ R, then $\displaystyle \sum_{n=1}^{10}$ f(n) is equal to

(A) 190 (B) 255 (C) 330 (D) 165

Ans. C

Sol. $f(1) = 3$

$f(2) = f(1)+f(1)+1 \quad 3 + 3 + 1 = 7$

$f(3) = f(2) + f(1) + 2 \quad 7 + 3 + 2 = 12$

$f(4) = 18$

$f(5) = 25$

$f(6) = 33$

$f(7) = 42$

$f(8) = 52$

$f(9) = 63$

$f(10) = 75$

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add = 330

Q25. If $f(x) = \displaystyle \frac{ax+b}{cx+d'}$, then $(fof)(x) = x$, provided that

(A) $d + a = 0$ (B) $d - a = 0$ (C) $a = b = c = d = 1$ (D) $a = b = 1$

Ans. A

Sol. $f(x) = \displaystyle \frac{ax\ +\ b}{cx\ +\ d} \Rightarrow fof(x) = \displaystyle \frac{a\bigg(\frac{ax\ +\ b}{cx\ +\ d}\bigg)+\ b}{c\bigg(\frac{ax\ +\ \ b}{cx\ +\ d}\bigg)+\ d} \Rightarrow fof(x) = \displaystyle \frac{a^2x\ +\ ab\ +\ bcx\ +\ bd}{acx\ +\ bc\ +\ cdx\ +\ d^2}$

fof (x) = $\displaystyle \frac{(a^2\ +\ bc)x\ +(ab\ +\ bd)}{(ac\ +\ cd)x\ +\ (bc\ +\ d^2)}$ = x

on comparing coefficient of both side $(a^2 + bc)x + (ab + bd)\ x^2 + (bc + d^2)x$

	$a^2 + bc = bc + d^2$	$\Rightarrow$	a = d or a = – d
and	ab + bd = 0	$\Rightarrow$	b = 0 or a = – d
and	ac + cd = 0	$\Rightarrow$	c = 0 or a = – d

Which can be simultaneously true for a = – d

Q26. Let $f(x) = x^2$ and g(x) = sin x for all x $\in$ R. Then the set of all x satisfying $(f\,\circ\,g\,\circ\,g\,\circ\,f)(x) = (g\,\circ\,g\,\circ\,f)(x)$, where $(f\,\circ\,g)(x)$ = $f(g(x))$, is

(A) $\displaystyle \pm \sqrt{n\pi},n \in \{0,1,2,......\}$ (B) $\displaystyle \pm \sqrt{n\pi},n \in \{1,2,......\}$

(C) $\displaystyle \frac{\pi}{2} +2{n\pi},n \in \{....-2,-1,0,1,2,......\}$

(D) $2{n\pi},n \in \{....,-2,-1,0,1,2,......\}$

Ans. A

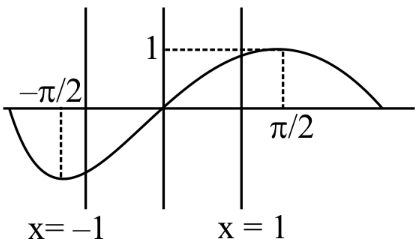
Sol. Given $f(x) = x^2$; g(x) = sinx

*f*ogogo f(x) = $\sin^2 (\sin x^2)$

and gogo f(x) = $\sin(\sin x^2)$

givne *f*ogogo f(x) = gogo f(x)

$\Rightarrow \sin^2(\sin x^2) = \sin(\sin x^2)$



$\Rightarrow \sin(\sin x^2)$ = 0 or 1 (rejected)

$\sin(\sin x^2) = 0 \qquad \Rightarrow x^2 = n\pi$

$\displaystyle \Rightarrow x = \pm \sqrt {n\pi } ;x \in \left\{ {0,1,2,3,.........} \right\}$

Q27. The function $f(x) = \log \displaystyle \bigg(\frac{1+\sin x}{1-\sin x}\bigg)$ is

(A) even (B) odd (C) neither even nor odd (D) both even & odd

Ans. B

Sol. $f(x) = \log \displaystyle \bigg(\frac{1 + \sin x}{1 - \sin x}\bigg) \Rightarrow f(-x) = \log \displaystyle \bigg(\frac{1 - \sin x}{1 + \sin x}\bigg) = -\log \displaystyle \bigg(\frac{1 + \sin x}{1 - \sin x}\bigg) = -f(x)$ odd function

Q28. The function $f(x) = [x] + \displaystyle \frac{1}{2}$, $x \notin I$ is a/an

(A) Even (B) odd (C) neither even nor odd (D) none of these

Ans. B

Sol. $f(x) = [x] + \displaystyle \frac{1}{2}$, $x \cancel{\in}$ I — $\cancel{\Rightarrow}$ $f[-x] + \displaystyle \frac{1}{2} = -[x] - 1 + \displaystyle \frac{1}{2} = -\displaystyle \bigg([x]+\frac{1}{2}\bigg) = -f(x)$ odd function

Q29. The inverse of the function $f(x) = y = \displaystyle \frac{e^x-e^{-x}}{e^x+e^{-x}}$ is

(A) $\displaystyle \frac{1}{2}\log\frac{1+x}{1-x}$ (B) $\displaystyle \frac{1}{2}\log\frac{2+x}{2-x}$

(C) $\displaystyle \frac{1}{2}\log\frac{1-x}{1+x}$ (D) $2 \log (1 + x)$

Ans. A

Sol. $\displaystyle \frac{y}{1} = \displaystyle \frac{e^x - e^{-x}}{e^x + e^{-x}}$

By componendo and dividendo

$\displaystyle \frac{1 + y}{1 - y} = \displaystyle \frac{2e^x}{2e^{-x}} \Rightarrow 2x = ln \displaystyle \bigg(\frac{1 + y}{1 - y}\bigg) \quad \therefore f^{-1}(x) = \displaystyle \frac{1}{2} ln \displaystyle \bigg(\frac{1 + x}{1 - x}\bigg)$

Q30. Fundamental period of f(x) = sec (sin x) is

(A) $\displaystyle \frac{\pi}{2}$ (B) 2π (C) π (D) aperidodic

Ans. C

Sol. f(x) = sec (sin x). Since sin x is a periodic function with fundamental period 2π. f(x) has a period 2π for fundamental period $f(x + \pi)$ = sec (sin $(\pi + x)$ = sec (– sin x) = sec (sin x) = f(x)

$f\displaystyle \bigg(x+\frac{\pi}{2}\bigg) \neq f(x)$ hence fundamental period is π

Q31. The fundamental period of function $f(x) = [x] + \displaystyle \bigg[x+\frac{1}{3}\bigg]+\bigg[x+\frac{2}{3}\bigg] - 3x + 15$

(A) $\displaystyle \frac{1}{3}$ (B) $\displaystyle \frac{2}{3}$ (C) 1 (D) non-periodic

Ans. A

Sol. $\displaystyle f\left(x \right) = \left[x \right] + \left[{x + \frac{1}{3}} \right] + \left[{x + \frac{2}{3}} \right] - 3x + 15$

$\displaystyle = \left\{ {\left[x \right] + \left[{x + \frac{1}{n}} \right] + \left[{x + \frac{2}{n}} \right] + ... + \left[{x + \frac{{n - 1}}{n}} \right] = \left[{nx} \right]} \right\}\,\,\,\,\,\,\,\,\,\,n \in N,x \in R$

$\displaystyle \Rightarrow f\left(x \right) = \left[x \right] + \left[{x + \frac{1}{3}} \right] + \left[{x + \frac{2}{3}} \right] = \left[{3x} \right]$

$\Rightarrow f(x) = [3x] - 3x + 15$

$= -(3x - [3x]) + 15$

$= -\{3x\} + 15$

Periodic function with period = $\displaystyle \frac{1}{3}$

Exercise J02

Q1. Read the symbols [ ] & { } as greatest integer function less than or equal to x & fractional part function..

(i) Find the number of real values of x, satisfying the equation $(x - 2)[x] = \{x\} - 1$.

(ii) Find the number of solutions of the equation, $x^2 - 3x + [x] = 0$ in the interval $[0, 3]$.

(iii) If $[x]^2 + 3[x] - 10 \geq 0$, then find the range of x.

(iv) If $[x]^2 - 5[x] + 6 = 0$, then find the range of x.

Ans. -

Sol. (i) $(x - 2)[x] = \{x\} - 1$

$\{x\} = x - [x]$

$\Rightarrow (x - 2)[x] = x - [x] - 1$

$\Rightarrow x[x] - 2[x] = x - [x] - 1$

$\Rightarrow [x](x - 1) = x - 1$

$\Rightarrow (x - 1)([x] - 1) = 0$

$x = 1$ or $[x] = 1$

$x = 1$ or $0 \leq x < 1$

$\Rightarrow 0 \leq x \leq 1$

Number of x satisfying given equation is infinite.

(ii) $x^2 - 3x + [x] = 0$; $x \in [0,3]$

If $0 \leq x < 1$ then $x^2 - 3x = 0 \Rightarrow x = 0$ or $x = 3$

$\Rightarrow x = 0$

If $1 \leq x < 2$ then $x^2 - 3x + 1 = 0 \Rightarrow x = \dfrac{3 \pm \sqrt{5}}{2}$

$\Rightarrow x \in f$

If $2 \leq x < 3$ then $x^2 - 3x + 2 = 0 \Rightarrow x = 2$ or 1

$\Rightarrow x = 2$

$\Rightarrow$ No. of values of x is 2.

(iii) $[x]^2 + 3[x] - 10 \geq 0$

let $[x] = t$

$t^2 + 3t - 10 \geq 0$

$\Rightarrow (t + 5)(t - 2) \geq 0$

$\Rightarrow$ 

$\Rightarrow t \in (-\infty, -5] \cup [2, \infty)$

$\Rightarrow [x] \leq -5$ or $[x] \geq 2$

$\Rightarrow x < -4$ or $x \geq 2$

$\Rightarrow x \in (-\infty, -4) \cup [2,\infty)$

(iv) $[x]^2 - 5[x] + 6 = 0$

let $[x] = t$

$\Rightarrow t^2 - 5t + 6 = 0$

$\Rightarrow (t-2)(t-3) = 0$

$\Rightarrow [x] = 2 \quad \text{or} \quad [x] = 3$

$\Rightarrow 2 \le x < 3 \quad \text{or} \quad 3 \le x < 4$

$\Rightarrow x \in [2,4)$

Q2. Find the domains of definitions of the following functions :
(Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

(i) $f(x) = \sqrt{\cos 2x} + \sqrt{16-x^2}$ (ii) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

(iii) $f(x) = \ln\left(\sqrt{x^2-5x-24-x-2}\right)$ (iv) $f(x) = \sqrt{\dfrac{1-5^x}{7^{-x}-7}}$

(v) $y = \log_{10}\sin(x-3) + \sqrt{16-x^2}$ (vi) $f(x) = \log100x \left(\dfrac{2\log_{10}+1}{-x}\right)$

(vii) $f(x) = \dfrac{1}{\sqrt{4x^2-1}} + \ln x(x^2-1)$ (viii) $f(x) = \sqrt{\log_{\frac{1}{2}} \dfrac{x}{x^2-1}}$

(ix) $f(x) = \sqrt{x^2|x|} + \dfrac{1}{\sqrt{9-x^2}}$ (x) $f(x) = \sqrt{(x^2-3x-10).\, ln^2(x-3)}$

(xi) $f(x) = \sqrt{\log_x(\cos 2\pi x)}$ (xii) $f(x) = \dfrac{\sqrt{\cos -(1/2)}}{\sqrt{6+35x-6x^2}}$

(xiii) $f(x) = \sqrt{\log_{1/3}(\log_4([x]^2-5))}$ (xiv) $f(x) = \dfrac{[x]}{2x-[x]}$

(xv) $f(x) = \log_x \sin x$

(xvi) $f(x) = \log_2\left(-\log_{1/2}\left(1 + \dfrac{1}{\sin\left(\frac{x^\circ}{100}\right)}\right)\right) + \sqrt{\log_{10}(log_{10}x) - \log_{10}(4-\log_{10}x) - \log_{10}3}$

(xvii) $f(x) = \dfrac{}{[x]} + \log_{1-\{x\}}(x^2-3x+10) + \dfrac{1}{\sqrt{2-|x|}} + \dfrac{1}{\sqrt{\sec(\sin x)}}$

(xviii) $f(x) = \sqrt{5x-6-x^2[\{ln\{x\}\}]} + \sqrt{7x-5-2x^2} + \left(ln\left(\dfrac{7}{2}-x\right)\right)^{-1}$

Ans. (i) $\left[-\dfrac{5\pi}{4}, \dfrac{-3\pi}{4}\right] \cup \left[-\dfrac{\pi}{4}, \dfrac{\pi}{4}\right] \cup \left[\dfrac{3x}{4}, \dfrac{5\pi}{4}\right]$ (ii) $\left(-4, \dfrac{1}{2}\right) \cup (2, \infty)$ (iii) $(-\infty, -3]$
(iv) $(-\infty, -1) \cup [0, \infty)$ (v) $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \le 4)$ (vi) $\left(0, \dfrac{1}{100}\right) \cup \left(\dfrac{1}{100}, \dfrac{1}{\sqrt{10}}\right)$
(vii) $(-1 < x < -1/2)$ U $(x > 1)$ (viii) $\left[\dfrac{1-\sqrt{5}}{2}, 0\right) \cup \left[\dfrac{1+\sqrt{5}}{2}, \infty\right)$ (ix) $(-3, -1]$ U $\{0\}$ U $[1,3)$

(x) $\{4\} \cup [5, \infty)$ (xi) $(0, 1/4) \cup (3/4, 1) \cup \{x : x \in N, x \geq 2\}$ (xii) $\left(-\frac{1}{6}, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 6\right)$

(xiii) $[-3, -2) \cup [3, 4)$ (xiv) $R - \left\{-\frac{1}{2}, 0\right\}$

(xv) $2K\pi < x < (2K + 1)\pi$ but $x \neq 1$ where K is non-negative integer

(xvi) $\{x \mid 1000 \leq x < 10000\}$ (xvii) $(-2, -1) \cup (-1, 0) \cup (1, 2)$ (xviii) $(1, 2) \cup (2, 5/2)$;

Sol. -

Q3. Find the domain & range of the following functions.

(Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

(i) $y = log_{\sqrt{5}}(\sqrt{2}\sin x - \cos x) + 3)$ (ii) $y = \frac{2x}{1+x^2}$ (iii) f(x) = $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

(iv) $f(x) = \frac{x}{1+|x|}$ (v) $y = \sqrt{2-x} + \sqrt{1+x}$

(vi) $f(x) = \log_{(\text{cosec } x - 1)}(2 - [\sin x] - [\sin x]^2)$ (vii) $f(x) = \frac{\sqrt{x+4} - 3}{x - 5}$

Ans. (i) D : $x \in R$ R : $[0, 2]$ (ii) D = R ; range $[-1, 1]$

(iii) D : $\{x \mid x \in R ;\ x \neq -3 ;\ x \neq 2\}$ R : $\{f(x) \mid f(x) \in R, f(x) \neq 1/5 ;\ f(x) \neq 1\}$

(iv) D : R ; R : $(-1, 1)$ (v) D : $-1 \leq x \leq 2$ R : $\left[\sqrt{3}, \sqrt{6}\right]$

(vi) D : $x \in (2n\pi, (2n+1)\pi) - \{2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{2}, 2n\pi + \frac{5\pi}{6} n \in I\}$ and

R : $\log_a 2$; $a \in (0, \infty) - \{1\} \Rightarrow$ Range is $(-\infty, \infty) - \{0\}$

(vii) D : $[-4, \infty) - \{5\}$; R : $\left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$

Sol. -

Q4. Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in N$ then find the value of $(a + b)$.

Ans. 5049

Sol. $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$

$x \in [-6, 6]$

$= (x^2 + 5x + 4)(x^2 + 5x + 6) + 5$

Let $t = x^2 + 5x$

$(t + 4)(t + 6) + 5 = t^2 + 10t + 24 + 5$

$= t^2 + 10t + 29$

$= (t + 5)^2 + 4$

$f(x) = (x^2 + 5x + 5)^2 + 4$

$4 \le f(x) \le (36 + 30 + 5)^2 + 4$

$4 \le f(x) \le 5041 + 4\ 5045$

$a + b = 5049$

Q5. Identify the pair(s) of functions which are identical?
(where [x] denotes greatest integer and {x} denotes fractional part function)
(i) $f(x) = \text{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$ (ii) $f(x) = \sqrt{\dfrac{1 - cos2x}{1 + \cos 2x}}$ and g (x) = tan x
(iii) $f(x) = \ln(1 + x) + \ln(1 - x)$ and $g(x) = \ln(1 - x^2)$ (iv) $f(x) = \dfrac{\cos x}{1 - \sin x}$ and $g(x) = \dfrac{1 + \sin x}{\cos x}$

Ans. (i), (iii) are identical

Sol. -

Q6. (i) Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :
(a) $10^x + 10^y = 10$ (b) $x + |y| = 2y$
(ii) The function f(x) is defined on the interval [0,1]. Find the domain of definition of the functions.
(a) f (sin x) (b) f (2x+3)
(iii) Given that y = f (x) is a function whose domain is [4, 7] & range is [–1, 9]. Find the range & domain of

(a) g (x) = f (x) (b) h (x) = f (x – 7)

Ans. (i) (a) $y = \log(10 - 10^x)$, $-\infty < x < 1$
(b) $y = x/3$ when $-\infty < x < 0$ & $y = x$ when $0 \le x < +\infty$
(ii) (a) $2K\pi \le x \le 2K\pi + \pi$ where () $K \in I$ (b) [–3/2 , –1]
(iii) (a) Range : [– 1/3, 3], Domain = [4, 7] ; (b) Range [–1, 9] and domain [11, 14]

Sol. -

Q7. Classify the following functions f(x) defined in $R \to R$ as injective, surjective, both or none .

(a) $f(x) = \dfrac{x^{+}4x + 30}{x^2 - 8x + 18}$ (b) $f(x) = x^3 - 6x^2 + 11x - 6$ (c) $f(x) = (x^2 + x + 5)(x^2 + x - 3)$

Ans. (a) neither surjective nor injective (b) surjective but not injective (c) neither injective nor surjective

Sol. -

Q8. Solve the following problems from (a) to (e) on functional equation.

(a) The function f (x) defined on the real numbers has the property that $f(f(x)) \cdot (1 + f(x)) = - f(x)$ for all x in the domain of f. If the number 3 is in the domain and range of f, compute the value of $f(3)$.

(b) Suppose f is a real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$. Find the value of $f(21)$.

(c) Let 'f' be a function defined from $R^+ \to R^+$. If $[f(xy)]^2 = x(f(y))^2$ for all positive numbers x and y and $f(2) = 6$, find the value of $f(50)$.

(d) Let f (x) be a function with two properties

(i) for any two real number x and y, $f(x + y) = x + f(y)$ and (ii) $f(0) = 2$.

Find the value of $f(100)$.

(e) Let f be a function such that $f(3) = 1$ and $f(3x) = x + f(3x - 3)$ for all x. Then find the value of $f(300)$.

(f) Suppose that f (x) is a function of the form $f(x) = \dfrac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$ $(x \neq 0)$.

If $f(5) = 2$ then find the value of $f(-5)$.

Ans. (a) – 3/4; (b) 64; (c) 30, (d) 102; (e) 5050; (f) 28

Sol. (a) $f(f(x))[1 + f(x)] = - f(x)$

$$f(f(x)) = \frac{-f(x)}{1 + f(x)} \quad \Rightarrow \quad f(x) = \frac{-x}{1 + x}$$

$$f(3) = \frac{-3}{1 + 3} = \frac{-3}{4}$$

(b) $f(x + f(x)) = 4f(x)$ & $f(1) = 4$

$f(1 + f(x)) = 4f(1) \Rightarrow f(1 + 4) = 16$

$f(5) = 16$

Now $f(5 + f(5)) = 4f(5)$

$f(5 + 16) = 64 = f(21)$

(c) $[f(xy)]^2 = x[f(y)]^2$ & $f(2) = 6$

put $x = 2$ & $y = 1$

$\Rightarrow [f(2\ 1)^2] = 2\ (f(1))2\ (f(1))^2 \Rightarrow (f(1))^2 = 18$

Now $f(50) = 30$

(d) $f(x + y) = x + f(y)$ & $f(0) = 2$

$f(100 + 0) = 100 + f(0) = 102$

(e) $f(3) = 1$

$f(3x) = x + f(3x - x)$

put $x = 1$

$f(3) = 1 + f(0)$

$f(0) = 0$

$f(6) = 2 + f(3) = 3$

$f(9) = 3 + f(6) = 3 + 3 = 6$

$f(12) = 4 + 6 = 10$

hence $f(300) = 1 + 3 + 6 + 1 + \ldots\ldots\ldots\ldots\ldots 100^{th}$ term

$S = 1 + 3 + 6 + 10 \ldots\ldots\ldots\ldots + T_n$

$S = 1 + 3 + 6 + \ldots\ldots\ldots + T_n$

$T_n = 1 + 2 + 3 + 4 \ldots\ldots\ldots$ up 100 term

$= \frac{100}{2} \quad 101 = 5050$

Q9. Suppose $f(x) = \sin x$ and $g(x) = 1 - \sqrt{x}$. Then find the domain and range of the following functions.

(a) fog (b) gof (c) fof (d) gog

Ans. (a) domain is $x \geq 0$; range $[-1, 1]$; (b) domain $2k\pi \leq x \leq 2k\pi + \pi$; range $[0, 1]$

(c) Domain $x \in R$; range $[-\sin 1, \sin 1]$; (d) domain is $0 \leq x \leq 1$; range is $[0, 1]$

Sol. -

Q10. A function $f : R \to R$ is such that $f\left(\frac{1-x}{1+x}\right) = x$ for all $x \neq -1$. Prove the following.

(a) $f(f(x)) = x$ (b) $f(1/x) = -f(x),\ x \neq 0$ (c) $f(-x - 2) = -f(x) - 2$.

Ans. -

Sol. -

Q11. $f(x) = \begin{bmatrix} 1-x & if & x \le 0 \\ x^2 & if & x > 0 \end{bmatrix}$ and $g(x) = \begin{bmatrix} -x & if & x < 1 \\ 1-x & if & x \ge 1 \end{bmatrix}$ find (fog)(x) and (gof)(x)

Ans. (gof)(x) = $\begin{bmatrix} x & if & x \le 0 \\ -x^2 & if & 0 < x < 1 \\ 1-x^2 & if & x \ge 1 \end{bmatrix}$; (fog)(x) = $\begin{bmatrix} x^2 & if & x < 0 \\ 1+x & if & 0 \le x < 1 \\ x & if & x \ge 1 \end{bmatrix}$

Sol. -

Q12. f (x) and g (x) are linear function such that for all x, $f(g(x))$ and $g(f(x))$ are Identity functions.

If f (0) = 4 and g (5) = 17, compute f (136).

Ans. 12

Sol. -

Q13. Find whether the following functions are even or odd or none

(a) f(x) = $\log\left(x + \sqrt{1+x^2}\right)$ (b) f(x) = $\frac{x(a^x+1)}{a^x-1}$ (c) f(x) = sin x + cos x

(d) f(x) = x $\sin^2$ x – x^3 (e) f(x)= sin x – cos x (f) f(x) = $\frac{(1+2^x)^2}{2^X}$

(g) f(x)= $\frac{x}{e^x-1} + \frac{x}{2} + 1$ (h) f(x) = $[(x+1)^2]^{1/3} + [(x-1)^2]^{1/3}$

Ans. (a) odd, (b) even, (c) neither odd nor even, (d) odd, (e) neither odd nor even, (f) even,

(g) even, (h) even

Sol. -

Q14. Compute the inverse of the functions:

(a) f (x) = ln $\left(x + \sqrt{x^2+1}\right)$ (b) f (x) = $2^{\frac{x}{x-1}}$ (c) y = $\frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

Ans. $(a)\frac{e^x - e^{-x}}{2}$; $(b)\frac{\log_2 x}{\log_2 x - 1}$; (c) $\frac{1}{2}\log\frac{1+x}{1-x}$

Sol. -

Q15. Function f and g are defined by f(x) = sin x, x ∈ R ; g(x) = tan x , x ∈ R - $\left(K + \frac{1}{2}\right)\pi$ where K ∈ I .

Find (i) periods of fog and gof. (ii) range of the function fog and gof.

Ans. -

Sol. fog(x) = f(tan x) = sin(tan x)

$$\Rightarrow \underbrace{fog\,(\pi + x) = fog\,(x)}_{P=\pi}$$

Range of fog is [–1,1]

$$\underbrace{gof(x) = g(\sin x) = \tan(\sin x)}_{\text{period}=2\pi}$$

Range = [–tan 1, tan 1]

Q16. Suppose that f is an even, periodic function with period 2, and that f (x) = x for all x in the interval [0, 1]. Find the value of f (3.14).

Ans. 0.86

Sol. -

Q17. The fundamental period of $e^{\cos^4}\pi x + x - [x] + \cos \pi x$ is ______. (where [ ] denotes the greatest integer function)

Ans. 2

Sol. -

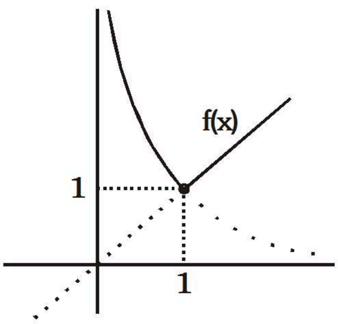
Q18. If f (x) = max $(x, 1/x)$ for x > 0 where max (a, b) denotes the greater of the two real numbers a and b. Define the function g(x) = f (x) · $f(1/x)$and plot its graph.

Ans. $\begin{bmatrix} \frac{1}{x^2} if 0 < x \le 1 \\ x^2 if x > 1 \end{bmatrix}$

Sol. $x = \frac{1}{x} \Rightarrow x^2 = 1 \quad \Rightarrow \quad x = 1$

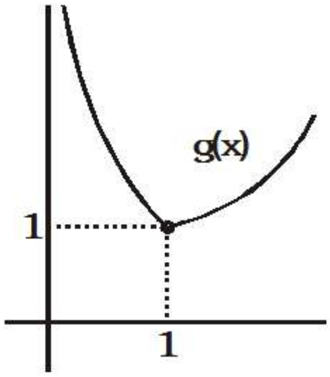
$f(x) = \frac{1}{x}; 0 < x \le 1$

$\quad = x \; ; x > 1$



$g(x) = f(x)\, f\left(\frac{1}{x}\right)$

$g(x) \begin{bmatrix} \frac{1}{x} \cdot \frac{1}{x} & 0 < x \le 1 \\ x\,x & x > 1 \\ & \end{bmatrix}$



Q19. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where $a, b, c, d \in R$. Suppose $P(0) = 6, P(1) = 7$, $P(2) = 8$ and $P(3) = 9$, then find the value of $P(4)$.

Ans. 34

Sol. -

Q20. Let $f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$. If $f(x)$ is divided by $x^3 - x$ then the remainder is some function of x say $g(x)$. Find the value of $g(10)$.

Ans. 21

Sol. -

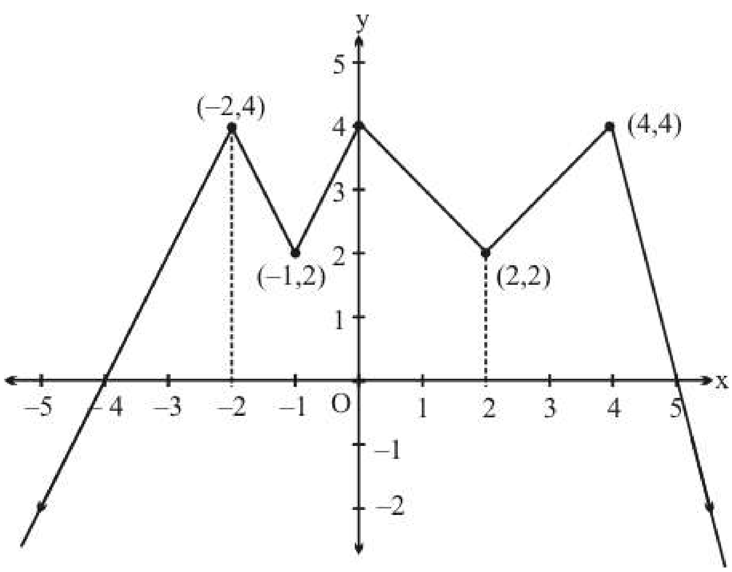
Q21. Let $f(x) = \dfrac{9^x}{9^x + 3}$ then find the value of the sum

$$f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \ldots\ldots\ldots + f\left(\frac{2005}{2006}\right)$$

Ans. 21

Sol. -

Q22. Consider the graph of a real-valued continuous function f(x) defined on R (the set of all real numbers) as shown below.



Find the number of real solutions of the equation $f(f(x)) = 4$.

Ans. 7

Sol. -

Paragraph

Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in R$

\n

Q23. Least value of 'a' for which f(x) is injective function, is

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

Ans. A

Sol. $f'(x) = x^2 + x + a \quad \Rightarrow \quad f'(x) \geq 0 \quad \Rightarrow \left(x+\frac{1}{2}\right)^2 + \left(a - \frac{1}{4}\right) \geq 0$

Hence $a \geq \frac{1}{4}$

Paragraph

Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in R$

\n

Q24. If $a = -1$ then f(x) is

(A) bijective (B) many–one and onto (C) one–one and into

(D) many– one and into

Ans. B

Sol. If $a = -1 \quad \Rightarrow \quad f(x) = x^2 + x - 1$

$D = 1 + 4 > 0 \quad \Rightarrow$ many one function and cubic polynomial have range $(-\infty, \infty)$ hece onto function

Paragraph

Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in R$

\n

Q25. f(x) is invertible iff

(A) $a \in \left[\frac{1}{4}, \infty\right), b \in R$ (B) $a \in \left[\frac{1}{8}, \infty\right), b \in R$ (C) $a \in \left(-\infty, \frac{1}{4}\right], b \in R$

(D) $a \in \left(-\infty, \frac{1}{4}\right), b \in R$

Ans. A

Sol. For f(x) to be invertible f(x) should be one-one that is f'(x) ≥ 0 $\Rightarrow$ $a \in \left[\frac{1}{4}, \infty\right)$

Paragraph

Consider the function $y = f(x) = x^2 - 6x + 5$

\n

Q26. The domain of definition of the function g (x) = $ln(f(x))$

(A) $(4, \infty)$ (B) $(-\infty, 1)$ (C) $\mathrm{R} - \{x|1 \leq x \leq 5\}$ (D) R

Ans. C

Sol. -

Paragraph

Consider the function $y = f(x) = x^2 - 6x + 5$

\n

Q27. Range of the function y = $ln(f(x))$ is

(A) R (B) $[1, \infty)$ (C) $[0, \infty)$ (D) $[-4, \infty)$

Ans. A

Sol. -

Paragraph

Consider the function $y = f(x) = x^2 - 6x + 5$

\n

Q28. Number of points of intersection of a line y = c (where c is a fixed constant and $0 < c < 4$)

with graphs $y = | f(|x|) |$ is

(A) 2 (B) 8 (C) 6 (D) 4

Ans. B

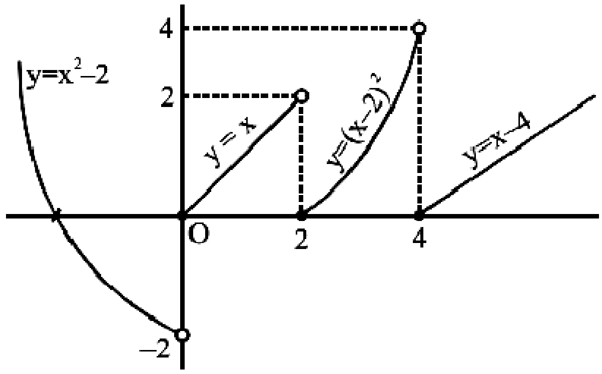
Sol. -

Q29. Consider $f(x) = \begin{cases} x^2 - 2 & x < 0 \\ x & 0 \leq x < 2 \\ (x-2)^2, & 2 \leq x < 4 \\ x - 4 & x \geq 4 \end{cases}$. If the equation f(x) = k has four roots, then k can be equal to :

(A) -2 (B) 0 (C) 1 (D) 2

Ans. B, C

Sol.



From graph obviously option (B) & (C) are correct.

Q30. Let $f(x) = x^2 + 4x - 1$ and $g(x) = |x|$. If $h(x) = f(g(x)) + 10$, then the range of $h(x)$, is subset of

(A) $\{y : y \geq 10\}$ (B) $\{y : y \geq 0\}$ (C) $\{y : y \geq 5\}$ (D) $\{y : y \geq 9\}$

Ans. B, C, D

Sol. $h(x) = f(g(x)) + 10 = |x|^2 + 4|x| - 1 + 10 = |x|^2 + 4|x| + 9 = (|x| + 2)^2 + 5$

Hence range of $h(x)$ is $[9, \infty)$

Minimum value occurs when $x = 0$]

Q31. Which of following pairs of functions are identical.

(A) $f(x) = e^{ln\sec^{-1}x}$ & $g(x) = \sec^{-1} x$ (B) $f(x) = \tan(\tan^{-1} x)$ & $g(x) = \cot(\cot^{-1} x)$

(C) $f(x) = \text{sgn}(x)$ & $g(x) = \text{sgn}(\text{sgn}(x))$

(D) $f(x) = \cot^2 x . \cos^2 x$ & $g(x) = \cot^2 x - \cos^2 x$

Ans. B, C, D

Sol. (A) $f(x) = e^{ln(sec^{-1}x)} = \sec^{-1} x,\ x\ (-\infty, -1] \cup (1, \infty)$

$g(x) = \sec^{-1} x,\ \ x\ (-\infty, -1] \cup [1, \infty)$

non-identical functions

(B) $f(x) = \tan(\tan^{-1} x) = x,\ x \in R$

$g(x) = \cot(\cot^{-1} x) = x,\ x \in R$

identical functions

(C) $f(x) = \text{sgn}(x) = \begin{cases} 1 & x < 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \Rightarrow g(x) = \text{sgn}(\text{sgn } x) = \begin{cases} 1 & x < 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

Identical functions

(D) $f(g) = \cot^2 x . \cos^2 x,\quad x \in R - \{n\pi\},\ n \in I$

$g(x) = \cot^2 x - \cos^2 x$

$= \cot^2 x\ (1 - \sin^2 x) = \cot^2 x.\ \cos^2 x \qquad x \in R - \{n\pi\},\ n \in I$

Identical functions

Q32. $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective.

(A) $f(x) = x^2$ (B) $g(x) = x^3$ (C) $h(x) = \sin 2x$ (D) $k(x) = \sin(\pi x/2)$

Ans. D

Sol. -

Q33. The functions which are aperiodic are (where [x] denotes greatest integer function)

(A) $y = [x + 1]$ (B) $y = \sin x^2$ (C) $y = \sin^2 x$ (D) $y = \sin^{-1} x$

Ans. D

Sol. -

Q34. If f: $R \to [-1, 1]$, where $f(x) = \sin\frac{\pi}{2}[x]$, (where [ ] denotes the greatest integer function) then:

(A) f (x) is onto (B) f (x) is into (C) f (x) is periodic (D) f (x) is many one

Ans. B, C, D

Sol.

$$f : R \to [-1, 1] \quad \Rightarrow \quad f(x) = \sin\left(\frac{\pi}{2}[x]\right) = \begin{cases} -1, & -1 \le x < 0 \\ 0, & 0 \le x < 1 \\ 1, & \text{i}1 \le x \le 2 \end{cases}$$

Many-one function

into function

Also $f(x + 4) = \sin\left(\frac{\pi}{2}[x+4]\right) = \sin\left(2\pi + \frac{\pi}{2}[x]\right) = \sin\left(\frac{\pi}{2}[x]\right)$

= f(x) and hece periodic

Q35. Identify the statement(s) which is/are incorrect ?

(A) the function $f(x) = \cos(\cos^{-1} x)$ is neither odd nor even

(B) the fundamental period of $f(x) = \cos(\sin x) + \cos(\cos x)$ is π

(C) the range of the function $f(x) = \cos(3 \sin x)$ is [- 1, 1] (D) none of these

Ans. A, B, C

Sol. -

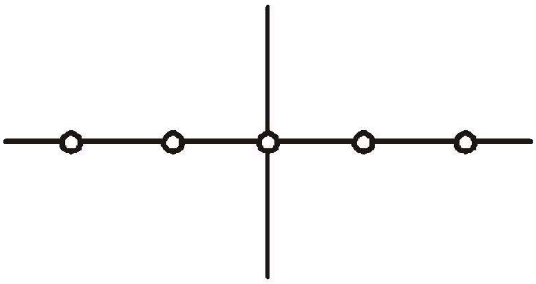
Q36. If F (x) = $\frac{\sin \pi[x]}{x}$, then F (x) is:

(A) periodic with fundamental period 1 (B) even (C) range is singleton

(D) identical to sgn $\left(sgn\frac{\{x\}}{\sqrt{\{x\}}}\right) - 1$, where {x} denotes fractional part function and [.] denotes greatest integer function and sgn (x) is a signum function.

Ans. A, B, C, D

Sol.



f(x) = $\frac{sin\ (\pi[x])}{[x]} = 0$, $x \notin I$

(B) By graph fundamental period is one

(B) f(– x) = 0 = f(x) $\therefore$ even function

(C) Range y $\in \{0\}$

(D) y = sgn $\left(sgn\frac{\{x\}}{\sqrt{\{x\}}}\right) -1$, $x \notin I$

y = sgn (1) – 1 $\Rightarrow$ y = 1 – 1

y = 0, $x \notin I$ Identical to f(x)

Q37. Function f(x) = sin x + tan x + sgn (x^2 – 6x + 10) is

(A) periodic with period 2π (B) periodic with period π (C) Non-periodic

(D) periodic with period 4π

Ans. A, D

Sol. -

Q38. Let f : R $\rightarrow$ R – {3} is a function satisfying f(x + α) = $\frac{f(x)-5}{f(x)-3}$ for all x $\in$ R, ten

(A) f is periodic with period 4α (B) f is into (C) f is constant

(D) f can not take the value 1

Ans. A, B, D

Sol. -

Q39. Let f(x) = ln x and g(x) = x2 – 1

Column-I contains composite functions and column-II contains their domain. Match the entries of column-I
with their corresponding answer is column-II.

Column-I	Column-II
(A) fog	(P) $(1, \infty)$
(B) gof	(Q) $(-\infty, \infty)$
(C) fof	(R) $(-\infty, -1) \cup (1, \infty)$
(D) gog	(S) $(0, \infty)$

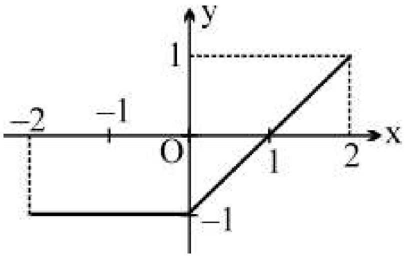
(A) (A) R (B) S (C) P (D) Q (B) (A) S (B) R (C) P (D) Q

(C) (A) R (B) S (C) Q (D) P (D) (A) R (B) P (C) S (D) Q

Ans. 5f0ea221b58d37a192817440

Sol. -

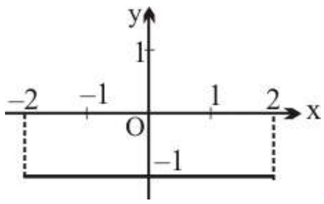
Q40. The graph of the function y = f (x) is as follows.



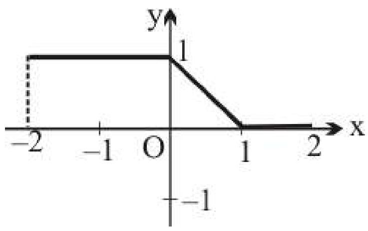
Match the function mentioned in Column-I with the respective graph given in Column-II.

Column-I Column-II

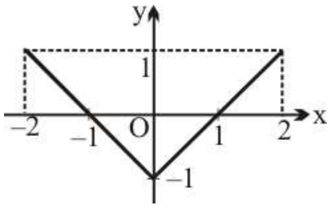
(A) y = | f (x) | (P)



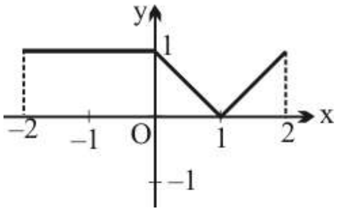
(B) y = f (| x |) (Q)



(C) $y = f(-|x|)$ (R)



(D) $y = \frac{1}{2}(|f(x)| - f(x))$ (S)



(A) (A) R (B) S (C) P (D) Q (B) (A) S (B) R (C) P (D) Q

(C) (A) S (B) P (C) R (D) Q (D) (A) Q (B) R (C) P (D) S

Ans. 5f0ea4f9b58d37a19281747d

Sol. -

Q41. Column I contains functions and column II contains their natural domains. Exactly one entry of column II matches with exactly one entry of column I.

Column I	Column II
(A) $f(x) = \sin^{-1}\left(\frac{x+1}{x}\right)$	(P) $(1, 3) \cup (3, \infty)$
(B) $g(x) = \sqrt{ln\left(\frac{x^2+3x-2}{x+1}\right)}$	(Q) $(-\infty, 2)$
(C) $h(x) = \frac{1}{ln\left(\frac{x-1}{2}\right)}$	(R) $\left(-\infty, -\frac{1}{2}\right]$
(D) $\phi(x) = ln\left(\sqrt{x^2+12} - 2x\right)$	(S) $[-3, -1) \cup [1, \infty)$

(A) (A) Q (B) S (C) P (D) R (B) (A) R (B) P (C) S (D) Q

(C) (A) R (B) S (C) P (D) Q (D) (A) S (B) R (C) Q (D) P

Ans. 5f0ea96a900175a12ca1d183

Sol. -